

## CHEMISTRY 101 · ANSWER KEY - PROFESSOR COPY

## Homework 2: Acids & Bases

Grading guide. Total 25 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

### Part A — Sort the kitchen

1. For each substance, write whether it is an acid, a base, or neutral — and whether it would taste sour, feel slippery, or neither. (1 point per row.) (6 pts)

Answer: Lemon juice — acid, sour. Baking soda water — base, slippery. Pure water — neutral, neither. Vinegar — acid, sour. Hand soap — base, slippery. Milk — (just barely) acid, mostly neither / a little sour when it goes off. 1 pt per row; half credit if one of the two words is right.

### Part B — The ten-times ruler

2. A drink is pH 6. Another is pH 4. How many times more acidic is the pH 4 drink? Show how you got it. (2 pts)

Answer: 100 times. Two steps down the ruler, and each step is ten times:  $10 \times 10 = 100$ . (1 pt for 100, 1 pt for the reasoning.)

3. Which is more acidic: pH 3 or pH 5? (1 pt)

Answer: pH 3 — lower number, more acid (100 times more, in fact).

4. Baking soda (a base) is stirred into vinegar (an acid) and it fizzes. What is happening? (1 pt)

Answer: A. The acid and base are neutralizing each other, and the fizz is  $\text{CO}_2$  escaping. - Acid + base cancel out. Bicarbonate catching the acid's loose hydrogens releases carbon dioxide — the fizz.

## Part C — Blood gas detective

*Normal values, about: pH 7.35-7.45;  $\text{CO}_2$  35-45; bicarbonate 22-26. Remember the two stories: pH low +  $\text{CO}_2$  high = the lungs aren't clearing acid (a breathing problem). pH low +  $\text{CO}_2$  normal + bicarbonate low = acid is coming from starving cells (find the cause). pH high +  $\text{CO}_2$  low = breathing too much.*

5. Four babies, four blood gases. For each, write what the problem is and what the team should do. (2 points per baby: 1 for the problem, 1 for the plan.) (8 pts)

Answer: A: normal — nothing to fix; keep watching. B:  $\text{CO}_2$  piling up, acid from the lungs not clearing (respiratory acidosis) — more breathing help: raise CPAP, ventilator breaths, caffeine. C: lungs are fine ( $\text{CO}_2$  normal) but the sponge is used up — acid from starving cells (metabolic acidosis) — hunt for the cause: infection, low blood pressure, sick gut; treat that. D: pH high,  $\text{CO}_2$  low — breathing too much, probably the ventilator is set too high — turn it down (low  $\text{CO}_2$  squeezes brain blood vessels).

## Part D — Flashback to Week 1 Chemistry

6. Carbonic acid is  $\text{H}_2\text{CO}_3$  and baking soda (sodium bicarbonate) is  $\text{NaHCO}_3$ . How many atoms are in each molecule? (Na is sodium — one atom.) (2 pts)

Answer:  $\text{H}_2\text{CO}_3$ :  $2 + 1 + 3 = 6$  atoms.  $\text{NaHCO}_3$ :  $1 + 1 + 1 + 3 = 6$  atoms. 1 pt each.

## Part E — Flashback to NICU Tutor

7. In the NICU, caffeine is a medicine. What is it for — and using today's chemistry, explain how it helps keep a preemie's blood pH steady. (2 pts)

Answer: Caffeine keeps the brain's breathing center awake, preventing apnea spells (1 pt). Steady breathing keeps blowing off  $\text{CO}_2$ , so carbonic acid doesn't pile up and the pH stays near 7.4 (1 pt).

## Part F — Think like a doctor

8. A preemie has a 25-second apnea spell with a brady. The nurse rubs her back and she restarts. Twenty minutes later a blood gas shows pH 7.24 and  $\text{CO}_2$  65. Explain what happened in her blood during the spell, and say two things the team might do next. (3 pts)

Answer: During the spell no  $\text{CO}_2$  was leaving, but every cell kept making it; it dissolved into carbonic acid and the pH slid down (1 pt). The gas shows the acid is from  $\text{CO}_2$  (high  $\text{CO}_2$ ), so it's a breathing problem, not starving cells (1 pt). Plans: check caffeine dose or start it, add or raise CPAP, watch for more spells and recheck a gas; don't over-treat one number in a baby who now looks fine (1 pt for any two sensible steps).

**HOME LAB (optional, not graded): The red-cabbage pH tester**

**You need:** a few red-cabbage leaves, hot water (grown-up helps), a jar, small cups, and things to test: lemon juice, vinegar, milk, plain water, baking-soda water, soapy water

1. Chop the cabbage, cover it with hot water, wait 10 minutes, and pour off the purple juice. That's your indicator.
2. Put a little of each test liquid in its own cup and add a spoonful of cabbage juice to each.
3. Watch the colors: pink or red means acid, blue or green means base, purple means neutral.
4. Line the cups up from reddest to greenest. That's a pH ruler. Write the order below, then try mixing the lemon cup into the baking-soda cup and watch the color (and the fizz).

**What you should see:** Lemon and vinegar turn pink/red (acids); baking-soda water and soapy water turn blue/green (bases); water stays purple; milk stays purple or goes very slightly pink. Mixing lemon into baking-soda water fizzes and the color heads back toward purple – neutralizing.